Henry Boyes

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SUMMARY

Machine learning engineer with production experience building end-to-end ML pipelines for large-scale scientific data. Designed and deployed a modular spatial proteomics pipeline processing ~3M cells using deep learning–based segmentation (Cellpose), unsupervised phenotyping (Scanpy/Leiden clustering), and Slurm-orchestrated GPU/CPU compute. Currently completing an M.S. in Data Science with coursework in machine learning, statistical analysis, and database systems. Strong background in systems thinking and process optimization from U.S. Navy service.

TECHNICAL SKILLS

Languages & ML: Python (scikit-learn, pandas, NumPy, Cellpose, Scanpy, AnnData, Squidpy), R, SQL
ML & Modeling: Supervised/unsupervised learning, deep learning inference, clustering (Leiden/UMAP), dimensionality reduction (PCA), spatial statistics, feature engineering
Pipeline & Infrastructure: Modular pipeline architecture, Slurm (HPC job orchestration), ETL workflows, GPU compute (A100), SQLite, AnnData/HDF5
Visualization & Reporting: matplotlib, seaborn, Tableau (learning), Power BI (learning)
Process Improvement: Lean Six Sigma Yellow Belt, project workflow optimization

PROFESSIONAL EXPERIENCE

Research Technician – Cleveland Clinic Foundation
(Mar 2025 – Present)

- Architected and implemented a production-grade spatial proteomics ML pipeline processing ~3M cells across multi-channel imaging datasets; pipeline stages span ingestion, preprocessing, deep learning segmentation (Cellpose v4), feature extraction, AnnData export, unsupervised phenotyping (Scanpy: arcsinh normalization, PCA, Leiden clustering, UMAP), and spatial analysis (Squidpy: neighborhood enrichment, co-occurrence, Ripley statistics).Manage a database of 700+ patient records with complex data structures, ensuring data quality, integrity, and compliance with standards while collaborating cross-functionally with clinical teams to support data-driven decision-making.
- Engineered Slurm orchestration scripts to distribute compute across GPU (A100, segmentation inference) and CPU (defq, downstream analysis) nodes, reducing manual intervention and enabling reproducible runs on HPC infrastructure.Collaborate cross-functionally with oncology researchers, clinical staff, and data management teams, providing analytical expertise and technical support that enables timely project completion and research advancement.
- Built automated data pipelines in Python (pandas, NumPy) that reduced data preparation time by 40%, processing large-scale datasets through ETL workflows to enable rapid analysis and standardized reporting.Perform ad hoc quantitative analysis to answer research questions, retrieving and manipulating data from multiple sources using SQL, Python, and applying statistical methods to validate hypotheses.

- Manage a clinical database of 700+ patient records with complex relational data structures; ensure data quality, integrity and compliance while supporting cross-functional oncology and clinical teams.
- Design and implement predictive models using scikit-learn and statistical methods to analyze high-dimensional clinical datasets, applying ML techniques for biomarker discover and treatment response prediction.

Research Assistant – Kent State University
(Aug 2023 – Oct 2024)
- Led the statistical analysis of material properties data resulting in publication in a high-impact journal.
- Designed analytical methodologies that improved data accuracy and processing efficiency, standardizing best practices across the research team.
- Presented research findings at conferences, translating complex data into accessible insights.

Hospital Corpsman 2nd Class, Quality Management – U.S. Navy, Naval Medical Center San Diego
(Dec 2018 – Apr 2020)
- Led data modernization project, transitioning from outdated legacy tracking systems to a new digital platform; improved HEDIS metrics by 60% through data-driven process redesign and performance monitoring.
- Analyzed performance data and designed optimization strategies that increased training readiness metrics from 60% → 95% within one month, demonstrating ability to drive measurable business outcomes.
- Supervised 2 junior staff; mentored and trained them, resulting in promotions.
- Recognized with Navy Achievement Medal and multiple performance awards.

Hospital Corpsman 3rd Class – U.S. Navy, Naval Medical Center San Diego
(Aug 2016 – Dec 2018)
- Maintained data accuracy and trained staff on documentation procedures, ensuring compliance with organizational standards.

Boatswain's Mate 3rd Class – U.S. Navy, USS Porter (DDG 78)
(Dec 2012 – Aug 2016)
- Managed project data and scheduling for 25+ personnel, tracking task completion and ensuring on-time delivery of mission-critical operations.

EDUCATION & CERTIFICATIONS

University of Pittsburgh (Expected 2027)
Master of Science – Data Science
Relevant Coursework - Machine Learning, Statistical Analysis, Database Systems**Kent State University** (May 2024)
Bachelor of Science – Organismal Biology
Cum laude, GPA: 3.6/4.0
Lean Six Sigma Yellow Belt (2019) | Project Management Performance Excellence (2019)

AWARDS

Navy Achievement Medal (×4), Navy Good Conduct Medal (×2) | Departmental Award for Significant Academic Achievement — Kent State University, 2024